STORE & INVENTORY MANAGEMENT MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · Product Team · School Administrators
Related Modules: Finance · Cafeteria · Assets · Procurement · Academics · Staff Management

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1. MODULE OVERVIEW

1.1 Purpose

The **Store & Inventory Management Module** manages all school inventory operations including:

- Food store supplies
- Office supplies
- Academic materials
- Laboratory equipment
- Uniform stock
- Maintenance supplies
- Asset consumables

The system ensures **efficient inventory tracking, transparency, cost control, and procurement automation**.

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2. CORE OBJECTIVES

The module must enable schools to:

1. Track **all inventory items**
2. Manage **stock levels**
3. Handle **stock requests**
4. Control **procurement workflows**
5. Monitor **stock movement**
6. Generate **inventory analytics**
7. Integrate **financial spending**
8. Prevent **stock shortages and wastage**

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3. SYSTEM ROLES & ACCESS CONTROL

Role	Permissions
Super Admin	Full platform oversight
School Admin	Inventory configuration
Store Manager	Manage inventory and approvals
Store Clerk	Stock updates and records
Department Head	Request inventory
Finance Officer	Procurement approvals
Auditor	Read-only access

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4. MODULE ARCHITECTURE

The **Store & Inventory Module** consists of the following subsystems:

```
Store & Inventory Management
│
├── Inventory Dashboard
├── Inventory Items
├── Stock Categories
├── Stock Requests
├── Stock Movements
├── Purchase Orders
├── Supplier Management
├── Department Allocation
├── Low Stock Alerts
├── Inventory Reports & Analytics
├── Audit Logs
```

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5. INVENTORY DASHBOARD

5.1 Overview

Provides administrators with **real-time visibility of inventory operations**.

Dashboard Widgets

- Total Inventory Items
- Low Stock Items
- Pending Requests
- Approved Purchases
- Inventory Value
- Recent Stock Movements

Example UI Layout

```
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Inventory Dashboard
----------------------------------------

Total Items: 1,240
Low Stock Alerts: 14
Pending Requests: 9
Purchase Orders: 6

Recent Movements:
+ 50 Exercise Books (Received)
- 10 Lab Gloves (Issued)
+ 20 Rice Bags (Received)
```

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6. INVENTORY ITEM MANAGEMENT

6.1 Overview

Stores all inventory records.

Each item contains:

- Identification
- Category
- Quantity
- Supplier
- Storage location
- Reorder level

6.2 Item Data Model

```
InventoryItem
- item_id
- item_name
- category_id
- quantity_available
- unit
- reorder_level
- supplier_id
- storage_location
- expiry_date
- created_at
```

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7. STOCK CATEGORIES

Allows classification of items.

Default Categories

- Food Store
- Office Supplies
- Laboratory Equipment
- Cleaning Supplies
- Uniforms
- Maintenance Tools

Example Structure

```
Categories
│
├── Food Store
│   ├── Rice
│   ├── Beans
│   ├── Cooking Oil
│
├── Office Supplies
│   ├── Paper
│   ├── Printer Ink
│   ├── Pens
```

Schools can **add custom categories**.

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8. STOCK REQUEST SYSTEM

Departments can request items from the store.

Example:

```
Science Department
Request:
- 20 Lab Gloves
- 10 Test Tubes
```

8.1 Request Workflow

```
Department Head
      ↓
Submit Request
      ↓
Store Manager Review
      ↓
Approval / Rejection
      ↓
Stock Issued
```

8.2 Request Data Model

```
StockRequest
- request_id
- department_id
- requested_by
- item_id
- quantity
- request_date
- status
```

Status types:

- Pending
- Approved
- Rejected
- Fulfilled

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9. STOCK MOVEMENTS

Tracks **all inventory transactions**.

Movement Types

- Stock Received
- Stock Issued
- Stock Adjustment
- Stock Transfer

9.1 Movement Data Model

```
StockMovement
- movement_id
- item_id
- movement_type
- quantity
- source
- destination
- performed_by
- timestamp
```

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10. PURCHASE ORDER MANAGEMENT

Handles **procurement processes**.

10.1 Procurement Workflow

```
Low Stock Alert
      ↓
Purchase Request
      ↓
Admin Approval
      ↓
Finance Approval
      ↓
Purchase Order Generated
      ↓
Supplier Delivery
      ↓
Stock Updated
```

10.2 Purchase Order Data Model

```
PurchaseOrder
- po_id
- supplier_id
- item_id
- quantity
- order_date
- delivery_date
- status
```

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11. SUPPLIER MANAGEMENT

Stores vendor information.

Supplier Details

- Supplier Name
- Contact Person
- Phone
- Email
- Address
- Product Types

Supplier Data Model

```
Supplier
- supplier_id
- supplier_name
- contact_person
- phone
- email
- address
```

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12. DEPARTMENT INVENTORY ALLOCATION

Tracks which department receives items.

Example:

```
Science Department
Received:
- 10 microscopes
- 50 lab gloves
```

Allocation Data Model

```
DepartmentInventory
- allocation_id
- department_id
- item_id
- quantity
- allocation_date
```

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13. LOW STOCK ALERT SYSTEM

The system monitors **reorder thresholds**.

Example:

```
Rice Stock: 5 bags
Reorder Level: 10

 Low Stock Alert
```

Alerts sent to:

- Store manager
- School admin

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14. INVENTORY REPORTS & ANALYTICS

Operational Reports

- Inventory stock report
- Department usage report
- Item consumption trends

Financial Reports

- Inventory valuation
- Procurement spending
- Supplier payments

Example Report

```
Monthly Consumption Report

Rice: 150kg
Beans: 100kg
Paper: 2000 sheets
```

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15. AUDIT TRAIL & SECURITY

Every action is logged.

Example:

```
AuditLog
- log_id
- user_id
- action
- item_id
- timestamp
```

Example Log:

```
Admin John
Updated quantity
Item: Rice
Old: 10 bags
New: 15 bags
```

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16. SYSTEM INTEGRATIONS

Module	Purpose
Finance	Procurement payments
Cafeteria	Food ingredient consumption
Assets	Equipment tracking
Staff	Department allocation
Academics	Lab equipment usage

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17. MOBILE INVENTORY SUPPORT

Staff can update inventory via mobile.

Features:

- Barcode scanning
- Quick stock updates
- Delivery confirmations
- Stock counts

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18. BARCODE / QR CODE SYSTEM

Each item can have a barcode.

Example workflow:

```
Scan Item
→ System identifies item
→ Update quantity
→ Save transaction
```

Benefits:

- Faster stock audits
- Reduced manual errors